The Half-Life of Compute: Modelling National AI Capacity Decay Under Supply Severance

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ABSTRACT

Export controls on artificial-intelligence hardware are justified by an implicit temporal claim: that restricting access to advanced accelerators buys time. The size of that interval has never been estimated. Existing measurement work treats national AI capacity as a *stock*: eleven public indices rank jurisdictions by installed compute, talent and infrastructure, and none models what becomes of that stock once resupply stops. Reliability research, conversely, measures accelerator failure under an explicit assumption of continuous replenishment. This paper joins the two. We model an installed base of AI accelerators as a perishable capital fleet subject to hardware failure, obsolescence against a moving capability frontier, partial salvage, and bounded domestic replacement, and we derive three reportable quantities: the compute half-life $T_{1/2}$, the frontier exit time T_f, and the sovereign floor C_∞. Failure rates are derived from published large-cluster telemetry (an annualised package failure rate of 9.08%, cross-validated against an independent estimate to within 1.4%). We obtain an analytic result: salvage absorbs the failure flow only above a critical cannibalisation ratio $\kappa^* = \lambda L$, equal to 0.38 under central parameters, beyond which further salvage capacity buys nothing and decline is set by obsolescence alone, while below it unrepaired failures accelerate decline by up to 40%. Because high-bandwidth memory is co-packaged with the die and neither is field-replaceable, the plausible range for accelerators straddles this threshold. Applying the model to three jurisdictions, we find frontier exit at 25-49 months, governed by $T_f \approx T\_dbl\ log_2(C_0/\theta\ F_0)$: installed stock buys time only **logarithmically**, so an order of magnitude more compute buys roughly 17 further months, and hardware decay accounts for only 13-14% of the interval. Median sovereign floors are 31.1% for the United States, 53.4% for China and 3.1% for the European Union; China's 90% interval, however, spans 22.1-155.7% of its base; whether China could sustain its fleet domestically is, on current public data, genuinely undetermined. Sensitivity analysis shows the rate of frontier advance dominates hardware attrition sevenfold. The policy implication is that severance need not destroy an adversary's compute; where remote access is also interdicted, freezing it suffices while the frontier moves.

Keywords compute governance; export controls; technological sovereignty; semiconductor supply chains; weaponized interdependence; AI policy; survival analysis

JEL F51, F52, L63, O33, O38 · **ACM CCS** Social and professional topics → Computing and business; Hardware → Reliability

1 Introduction

Every argument for controlling the export of artificial-intelligence hardware rests on a claim about time. Restricting an adversary's access to advanced accelerators is said to *buy* something: a lead, a window, a delay before capability parity. The claim is temporal, quantitative and, so far as we have been able to establish, unquantified. No published work states how long a jurisdiction cut off from the global accelerator supply chain could sustain the compute base it already owns.

This is not for want of measurement. National artificial-intelligence capacity is now one of the most heavily indexed quantities in technology policy. At least eleven public instruments rank jurisdictions on some combination of installed compute, energy, talent, model output and governance capacity. They differ in construction and in candour, but they share a structural property: each is a **snapshot of holdings**. They answer *what does a country have?* None answers *what would a country keep?*

The distinction matters because installed compute is not a reserve. It is a capital fleet with a measurable failure rate, a contested economic life, a dependence on a separately concentrated memory supply, and a capability that is defined relative to a frontier that does not stand still. A stock of gold does not corrode; a stock of accelerators does. Treating the two alike is the central modelling error this paper sets out to correct.

The reliability literature has measured the corrosion rate precisely, but under an assumption that voids its use here. Large-cluster telemetry gives well-characterised accelerator failure statistics, and the leading analysis of whether such failures constrain scaling concludes that they do not, while stating plainly that it assumes replacement nodes remain procurable and does not address supply-chain disruption or situations in which replacement chips cannot be obtained. That assumption is exactly what severance removes. Relaxing it is the contribution of this paper.

1.1 Contribution

1. **A decay model of installed compute under severance.** We treat a national accelerator base as a perishable fleet subject to failure, obsolescence, partial salvage and bounded domestic replacement, and integrate it forward monthly. Failure parameters are derived from published telemetry rather than assumed, and cross-validated against an independent estimate.
2. **Three reportable quantities.** The compute half-life $T_{1/2}$, the frontier exit time T_f, and the sovereign floor C_∞, the capacity a jurisdiction retains asymptotically from domestic production alone. C_∞ is, we argue, the quantity that the word *sovereignty* has been used to gesture at without ever being given a number.
3. **An analytic threshold.** Salvage sustains an installed base only above a critical cannibalisation ratio $\kappa^* = \lambda L$. This is a property of the hardware, not of policy, and it separates two qualitatively different severance regimes.
4. **A negative result reported in full.** We attempted to calibrate the model against the only observed severance of a comparable capital fleet, Russian civil aviation after February 2022, and the calibration fails. We report the failure, diagnose it, and explain why it constrains the analogy rather than the model.

We deliberately do not propose another index. The field has eleven, and an eleventh-and-a-half would be neither novel nor useful. Existing indices enter this paper as inputs and related work, not as rivals.

2 Related work

2.1 Measuring national technological capacity

Composite indices of technological capacity are numerous and methodologically mature. The Machinepower Index (2026) scores twenty-five jurisdictions across twelve cells grouped as Watts, Weights and Will, aggregating with a constant-elasticity-of-substitution mean at $\sigma = 0.33$ so that a single weak cell constrains the total rather than being averaged away; its own documentation records that 37% of underlying measurements are analyst judgement, and that no time dimension, decay, depreciation, failure rate or severance scenario is modelled. The CNAS Sovereign AI Index (2026) catalogues over 139 state-backed projects and finds that most remain dependent on foreign , predominantly United States, technology across the stack; it is explicitly not a dependency-severance model. Academic instruments are similar in structure: Lee et al. (2024) decompose technology sovereignty into innovation capability, production capability and supply-chain independence for the semiconductor industry; Caravella et al. (2023) trace strategic dependencies through the photovoltaic supply chain; Cai et al. (2026) review 104 articles on semiconductor industrial policy and regional ecosystem governance. Broader instruments, the Government AI Readiness Index, the Global AI Index, the annual AI Index, extend coverage without changing the temporal logic.

Every one of these is cross-sectional. The methodological literature on composite indicators is itself explicit that weighting, aggregation and robustness are where such instruments succeed or fail (Greco et al., 2018; Dobbie et al., 2013; Kelemen et al., 2024; OECD/JRC, 2008), and we adopt its discipline below. But no amount of care in aggregation converts a snapshot into a trajectory.

2.2 Compute governance and accelerator reliability

Sastry et al. (2024) establish the governing framework: computing power is a tractable object of policy because it is detectable, excludable, quantifiable and produced through an extremely concentrated supply chain. Those four properties are precisely the conditions under which a severance model is possible; this paper can be read as the quantitative successor to that qualitative argument. Regulatory practice has moved the same way, with cumulative processing performance rather than unit counts becoming the controlled quantity.

On the hardware side, Grattafiori et al. (2024) report component-resolved interruption statistics for a 16,384-accelerator pre-training run, and Kokolis et al. (2025) fit failure models across more than 150 million accelerator-hours spanning two production clusters. Epoch AI's analysis concludes that hardware failure will not limit scaling, under continuous replenishment. Our reading of that boundary condition, stated in the source itself, motivates the present work.

A third lineage is closer than either and must be acknowledged: military sustainment modelling. Sherbrooke's METRIC (1968) and the multi-echelon tradition it founded optimise recoverable-spare stocks for fleet availability, and its modern extensions incorporate cannibalisation explicitly. That literature is the nearest quantitative relative of our salvage term, and we borrow its vocabulary. It differs on exactly the margins this paper needs: it optimises inventory under a functioning resupply pipeline rather than total severance, and its systems face no moving capability frontier, an aircraft that flies is available, whereas an accelerator that runs may already be strategically obsolete. The threshold κ^* should accordingly be read as the severance-regime limit of that tradition, not as a claim to have discovered cannibalisation.

2.3 Economic statecraft and chokepoints

Farrell and Newman (2019) supply the theoretical frame: asymmetric network topology allows states with jurisdiction over central nodes to exercise a *panopticon effect* and a *chokepoint effect*, the latter denying network access to adversaries. Beaumier et al. (2023) apply network analysis directly to semiconductors, decomposing the supply chain into design, raw material, manufacturing-equipment and assembled-chip networks and showing how centrality in one enables weaponisation of another. Fuller (2026) argues the framework transfers imperfectly to physical goods, where a chokepoint is better understood as an input without which a task cannot continue, a formulation that maps closely onto the model developed here.

Quantitative evaluation of controls has concentrated on economic aggregates. Park and Liu (2023) use multi-regional input-output methods; Cui et al. (2025) employ a dynamic computable general equilibrium model to estimate GDP effects of the chip embargo and of Chinese counter-controls on gallium, germanium and graphite; Shrivastava et al. (2025) assess circumvention. Supply-chain resilience has been modelled with scale-free cascade topologies and Bayesian networks. All of this measures **economic** consequence. None measures **capability decay over time**.

Industry analysis comes closest. Estimates of when a stockpile of unassembled dies will be exhausted have been published, including a zero-inflated formulation assigning roughly a 56% probability that one such stockpile was depleted by January 2026. That work models *inventory drawdown*, components awaiting assembly. The present paper models *installed-base decay*, hardware already deployed and running. The two are different questions with different mathematics, and we are careful to claim only the second.

> *Prior work measures what a jurisdiction holds, what it can build, and what severance costs its economy. This paper measures how long what it holds keeps working.*

3 Model

3.1 Setup

Let $N(t)$ denote the number of operational accelerators, in H100-equivalents (H100e), at month t after severance, with $N(0) = C_0$. Effective national compute is

$$C(t) \;=\; N(t)\,u(t) \tag{1}$$

where $u(t) \in (0,1]$ is a utilisation ceiling imposed by energy, interconnect and operational constraints. Three flows act on N each period. Hardware failures occur at monthly hazard $\lambda = -\ln(1\text{-}AFR)/12$. Obsolescence retires units at rate $\delta = 1/L$, with L the useful life in months; retired units remain physically intact. Replacement arrives as domestic production plus leakage, $R = R\_dom + R\_leak$.

The mechanism that distinguishes severance from normal operation is salvage. Under resupply a failed unit is replaced and N is unchanged, which is why reliability analyses conclude that failures do not bind. Under severance the only source of repair is other units. Let $P(t)$ be the pool of salvageable repairs and let κ be the *cannibalisation yield*: the number of repairs recoverable from one donor unit. Then

$$F_t = N_t\lambda, \qquad \mathrm{Ret}_t = N_t\delta$$

$$\rho_t = \min(F_t,\ P_t)$$

$$P_{t+1} = P_t - \rho_t + \kappa\,(F_t - \rho_t + \mathrm{Ret}_t)$$

$$N_{t+1} = N_t - F_t - \mathrm{Ret}_t - S_t + \rho_t + R \tag{2}$$

where S_t is the number of *serviceable* units voluntarily withdrawn for parts. Withdrawal is rational only when one donor yields more than one repair, so $S_t > 0$ requires $\kappa > 1$; we cap it at 3% of the fleet per month.

3.2 The critical cannibalisation ratio

The salvage pool is replenished by units leaving service and drained by repairs. Retirements alone sustain the repair flow when $\kappa N\delta \geq N\lambda$, which gives a threshold independent of fleet size, inflow and policy:

$$\kappa^{*} \;=\; \frac{\lambda}{\delta} \;=\; \lambda L \tag{3}$$

κ^{*} is a **saturation** threshold, not a survival threshold, and the distinction matters. Above it, salvage absorbs essentially the entire failure flow and the installed base declines at the obsolescence rate alone; further salvage capacity buys nothing. Below it, unrepaired failures accumulate on top of retirement, and at $\kappa = 0$ the fleet contracts 40% faster than obsolescence alone would imply. What κ^{*} separates is a *failure-limited* regime from an *obsolescence-limited* one. It does not separate survival from collapse: absent replacement inflow every fleet declines, because retirement removes units whether or not failures are repaired. κ^{*} is a property of the hardware, its failure rate and its useful life, not of the jurisdiction holding it. Appendix A gives the derivation; §5.1 gives the verification.

This threshold is where the physical architecture of accelerators becomes decisive, and where the analogy with other capital fleets breaks. An airframe is parts-rich: it decomposes into thousands of independently failing line-replaceable units, and observed practice under sanctions has been to dismantle roughly one airframe to keep about ten flying, implying $\kappa \approx 10$. An AI accelerator is parts-poor in exactly the wrong place. High-bandwidth memory is co-packaged with the logic die on a shared interposer; neither the die nor the memory stack is a field-replaceable unit. The two dominant failure modes are therefore non-salvageable **by construction**, and cannibalisation recovers only ancillary node components, power supplies, network interfaces, cooling, which are not the binding failure modes. We take κ □ $[0.05, 0.50]$ for accelerators and treat the aviation value as an upper bound belonging to a different hardware class.

3.3 Reported quantities

- **Compute half-life** $T_{1/2}$, months until effective compute falls below half its severance-day value. Right-censored at the 120-month horizon.
- **Frontier exit time** T_f, months until national capacity falls below a threshold share $\theta = 0.10$ of the contemporaneous frontier training scale $F(t) = F_0 \ 2$^t/T_dbl, where F_0 is an **absolute, jurisdiction-common** anchor. An earlier draft normalised by each jurisdiction's own C_0 instead, which forced every jurisdiction to begin at parity and made C_0 cancel identically; the apparent stock-invariance that followed was an algebraic artefact and has been withdrawn (§5.3, Appendix A).
- **Sovereign floor** C_∞, the asymptotic level as a share of C_0, sustainable from all non-severed inflow, domestic production *and* residual leakage, held at its severance-date rate. A closed form exists and is given in Appendix A; it reproduces the 900-month integration exactly. C_∞ is defined only for zero indigenisation growth (§7.4).

The anchor F_0 deserves its derivation in the open, since T_f inherits it through the logarithm. A frontier training run of 10^{26} to 10^{27} FLOP, executed over roughly three months on accelerators sustaining an effective 4×10^{14} FLOP/s, corresponds to about 3×10^{21} FLOP per accelerator and hence to 3×10^4 to 3×10^5 H100e per run. Live estimates bracket the same interval: the first disclosed run above 10^{26} FLOP was Grok 3 in February 2025 at roughly 3×10^{26}, the largest run to date is Grok 4, and Epoch AI places the top frontier model's crossing of 10^{26} around January 2026. We therefore sweep F_0 log-uniformly over 5×10^4 to 5×10^5 H100e with a central value of 1.5×10^5, and every reported T_f carries that uncertainty.

4 Data and parameters

4.1 Failure rates derived from cluster telemetry

Grattafiori et al. (2024) report 419 unexpected interruptions over a 54-day pre-training window on a cluster of up to 16,384 H100 accelerators, resolved by cause: faulty accelerator 148 (30.1%), accelerator HBM3 memory 72 (17.2%), network switch or cable 35 (8.4%). Annualising per accelerator,

$$\mathrm{AFR} = \frac{n_{\mathrm{events}}}{16{,}384} \times \frac{365}{54} \tag{4}$$

gives 6.11% per year for die failures and 2.97% for HBM3, hence a **package failure rate of 9.08% per year** for the two non-field-replaceable modes, and 17.29% across all interruption causes. As an independent check, the all-cause figure corresponds to one interruption per 50,677 accelerator-hours, against roughly 50,000 reported by Epoch AI on the same underlying run, agreement to within 1.4%. We treat the package rate as the central value and sweep the full interval in the Monte Carlo. One caution attaches to these rates: interruption counts do not distinguish terminal hardware death from transient faults cleared by a reset, so to the extent some die-attributed interruptions were recoverable, the package rate overstates permanent attrition. The direction of that bias is benign for our conclusions: a lower λ lengthens $T_{1/2}$, raises floors and leaves T_f nearly unchanged.

4.2 Installed stock

Epoch AI estimates a global delivered stock of approximately 15 million H100e and publishes ownership-resolved data. For China we derive C_0 from a published relation rather than an assumption: the median estimate of 660,000 H100e diverted through 2025 is characterised as roughly one third of Chinese total capacity, giving $C_0 \approx 1{,}980{,}000$ H100e. The 90% interval on the smuggling estimate (290,000-1,600,000) is propagated directly.

This is the study's principal data limitation and we state it plainly. Epoch resolves ownership by *firm*, not by *jurisdiction*. Attributing a multinational operator's fleet to a country requires assumptions that are not fully defensible from public data. We therefore sweep the United States share over 60-78% and the European Union share over 2-6% of global stock, and we report European results with correspondingly low confidence. No conclusion in this paper depends on a point estimate of national stock.

4.3 Severance is modelled symmetrically

A jurisdiction is severed from the inputs it does not control. It is worth stating that this is not a China-specific scenario. The United States has no domestic source of extreme-ultraviolet lithography, holds only partial and still-ramping leading-edge fabrication capacity, and has no high-bandwidth memory production at scale, HBM supply is roughly 90% concentrated in two Korean firms. American domestic coverage under severance from its own allies is therefore a scenario band (15-35%), not unity. The European Union, with no leading-edge logic fabrication in operation, is assigned 1-4%.

Table 1. Model parameters, ranges and sources.

Symbol	Parameter	Central	Range	Source
λ	Package failure rate (per yr)	9.08%	6.11-17.29%	Derived, Grattafiori et al. (2024)
κ	Cannibalisation yield	0.275	0.05-0.50	Package architecture (§3.2)
L	Useful life (yr)	4.0	2.0-6.0	Depreciation dispute, 2025-26
T_dbl	Frontier doubling (mo)	5.2	4.8-7.0	Epoch AI, frontier *training* series
F_0	Frontier anchor (H100e)	150,000	50k-500k	Derived; see §3.3. Swept log-uniform
θ	Competitiveness threshold	0.10	fixed	Convention; enters via the logarithm (§5.3)
cov	US / EU domestic coverage	0.25 / 0.02	0.15-0.35 / 0.01-0.04	**Assumption, no observational basis**
C_0^CN	China stock (H100e)	1,980,000	0.87-4.80 M	Derived, Epoch AI
C_0^US	US stock (H100e)	10,500,000	9.0-11.7 M	Epoch AI, share swept
C_0^EU	EU stock (H100e)	600,000	0.3-0.9 M	Scenario, low confidence
R_dom	China domestic (H100e/mo)	28,708	30-100% of 2025 rate	Epoch AI realised output × stockpile dependence
R_leak	Leakage (H100e/mo)	9,625	10-60% of base	Derived, Epoch AI (2026)

H100e = H100-equivalents. Ranges are propagated through a 10,000-draw Monte Carlo with independent uniform sampling. Independence is itself an assumption: plausible correlations among parameters, between useful life and failure rate, or between stock and domestic capacity, are not modelled.

5 Results

5.1 The plausible cannibalisation range straddles the threshold

Under central parameters $\kappa^* = 0.38$. Across the parameter grid it ranges from 0.13 (die failures only, two-year life) to 1.14 (all-cause failures, six-year life). The plausible range for accelerators, *[0.05, 0.50]*, **straddles the central threshold**. Direct measurement of the integrated model confirms the algebra: against a retirement-only decline of 2.08% per month, the fleet contracts at 2.92% per month at $\kappa = 0$ (40% excess), falling to 2.14% at κ^* (3% excess), after which tripling κ changes it by less than 2%. A fleet below the threshold therefore pays a compounding penalty, and one above it gains nothing from further salvage capability. Establishing κ empirically is, on this analysis, the highest-value measurement the field could undertake.

The threshold interacts with the depreciation dispute in a way worth stating. Because $\kappa^* = \lambda L$ rises with useful life, a longer-lived fleet sits further below any fixed salvage capability: across the disputed two-to-six-year range the share of failures salvage can absorb falls from 100% to 48%. We tested, and reject, the tempting inference that longer-lived fleets are therefore more fragile overall: time-to-half under zero inflow *rises* from 17 to 41 months across the same range, because slower retirement dominates. The defensible statement is narrower, a six-year fleet survives longer in absolute terms while depending more on raw stock and less on repair. We report the rejected hypothesis because it is attractive, quotable and false.

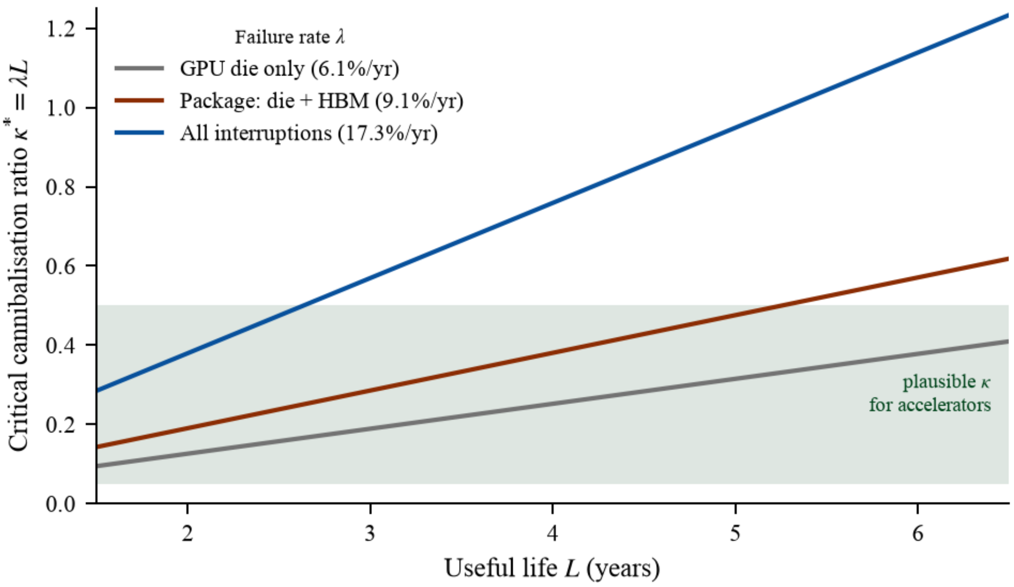

Figure 1. The critical cannibalisation ratio $\kappa^* = \lambda L$ as a function of useful life, for three failure-rate assumptions. The shaded band is the plausible range for AI accelerators given that high-bandwidth memory is co-packaged and non-field-replaceable. The band intersects all three curves: whether salvage sustains an installed base is regime-dependent, not a matter of degree.

5.2 Trajectories and reported quantities

Table 2 reports medians and 90% intervals from 10,000 draws per jurisdiction. Frontier exit occurs between 25 and 49 months, ordered by installed stock as the analytic relation in §5.3 requires. At the median no jurisdiction reaches self-sufficiency, but the floors must be read with double care, for the United States and European Union they are the assumed domestic-coverage parameter rescaled by a bounded factor, not independent model output (§7.2), while China's 90% interval reaches 155.7%, above parity. Median sovereign floors are 31.1% for the United States, 53.4% for China and 3.1% for the European Union.

Table 2. Reported quantities by jurisdiction. Medians with 90% intervals, 10,000 Monte Carlo draws.

Jurisdiction	C_0 (M H100e)	$T_{1/2}$ (mo)	T_f (mo)	C_∞ (% of C_0)	$P(T_{1/2} \leq 36)$
China	1.98	>120 [29->120]	41 [30-54]	53.4 [22.1-155.7]	11.3%
United States	10.50	51 [28-111]	49 [38-63]	31.1 [19.5-44.7]	19.2%
European Union	0.60	28 [18-43]	25 [17-36]	3.1 [1.4-5.0]	82.5%

$T_{1/2}$ is right-censored at 120 months. Censored draws are retained ordinally, so any quantile falling in the censored mass is reported as '>120' rather than as a finite value; censoring affects 57.7% of Chinese draws and 3.8% of US draws. European figures carry low confidence because jurisdiction-resolved stock data are weakest (§4.2).

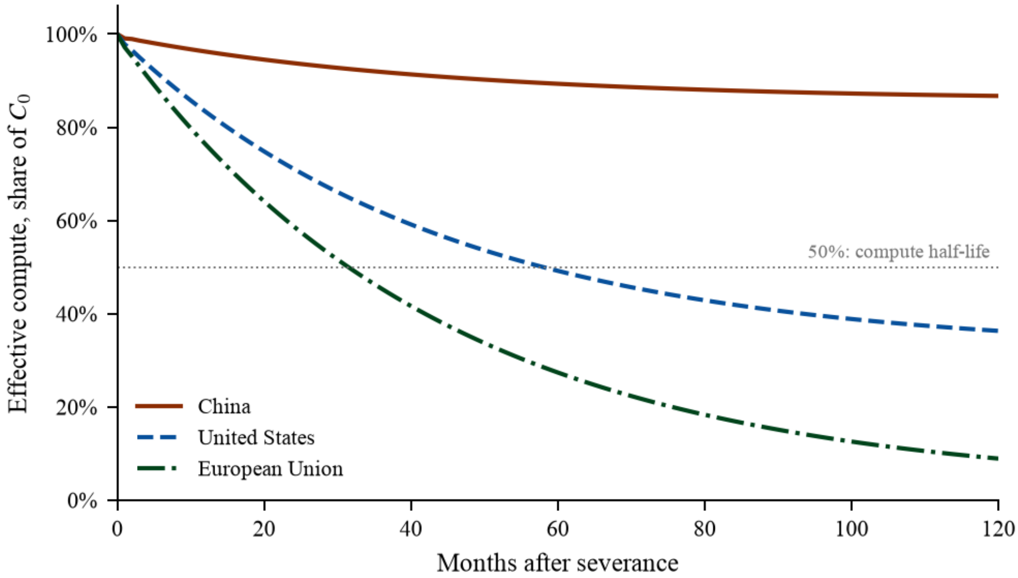

Figure 2. Central-case decay of effective compute after severance. The European trajectory falls fastest despite the smallest absolute exposure, because domestic replacement covers the least of its attrition.

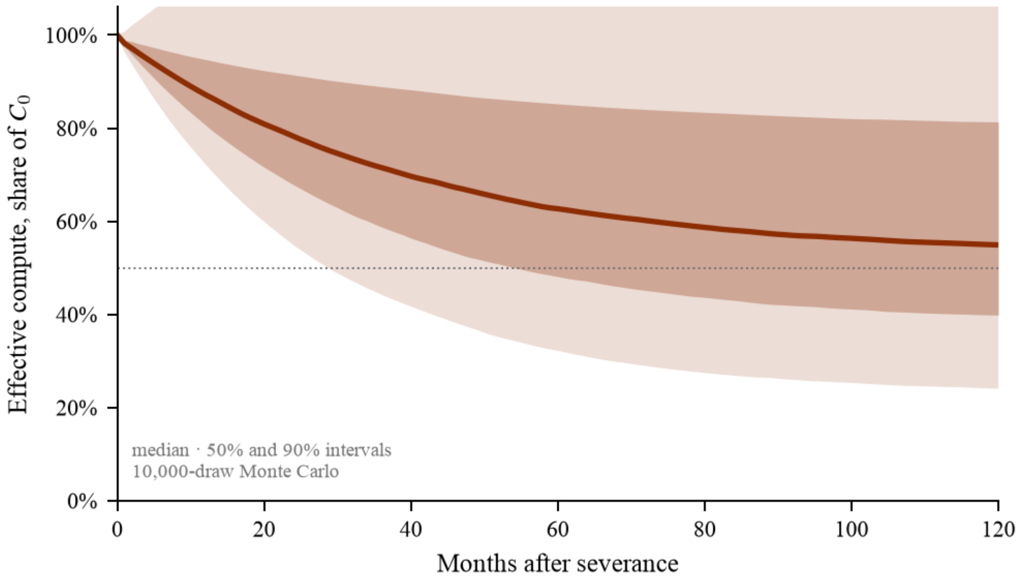

Figure 3. Monte Carlo distribution for China, 10,000 draws with independent uniform sampling over all parameters in Table 1. The width of the 90% interval reflects the joint sweep of all parameters; because China's inflows are absolute rather than proportional to stock, uncertainty in C_0 contributes as well.

5.3 The frontier clock dominates hardware attrition

The consequential result is not the decay of hardware but its small share of the interval. Figure 4 plots capability against the contemporaneous frontier scale. Setting decay aside entirely, the exit time follows

$$T_f \simeq T_{\text{dbl}} \log_2 \left(\frac{C_0}{\theta F_0} \right) \tag{5}$$

which reproduces the simulated central cases to within four months (analytic 36.6 / 49.1 / 27.7 against simulated 36 / 45 / 24 for China, the United States and the European Union). Two consequences follow, and they replace the stock-invariance claim made in an earlier draft.

- **Installed stock buys time only logarithmically.** At $T\_dbl = 5.2$ months, a tenfold larger fleet buys $T\_dbl \log_2 10 \approx 17$ further months, and a hundredfold fleet only 35. Stockpiling is subject to sharply diminishing returns against a moving frontier, a stronger and more useful statement than the invariance claim it replaces, and unlike that claim it is not an artefact of normalisation.

- **Hardware decay accounts for a consistent minority of the interval.** Re-running each central case with failure and retirement set to zero gives 42 / 52 / 28 months against 36 / 45 / 24 with decay: attrition costs 6 / 7 / 4 months, or **13-14% of the budget** in all three jurisdictions. The remaining 86% is the frontier clock.

The corrected reading is therefore narrower than the original but survives scrutiny: hardware reliability is not what governs strategic position under severance, but neither is position independent of how much hardware a jurisdiction holds.

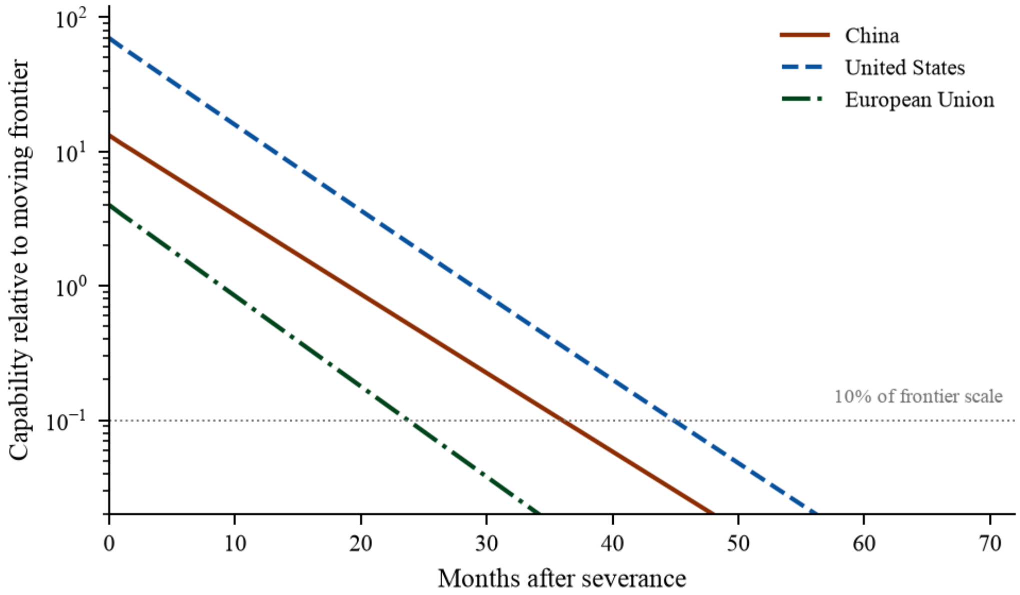

Figure 4. Capability against an absolute frontier anchor, logarithmic scale. Starting positions differ because installed stocks differ; the near-parallel decline reflects the common frontier deflator. Crossing times are ordered by stock and separated by roughly $T\_dbl \log_2$ of the stock ratio.

Sensitivity analysis confirms the mechanism. Varying the frontier doubling time across its 4.8-7.0 month range moves T_f by 15 months; varying the failure rate across its full derived interval moves it by 2 months, a sevenfold difference in leverage. Hardware reliability, the quantity the engineering literature measures most carefully, is among the *least* important parameters for the strategic outcome. This is the one place where the correction to *T_dbl* strengthened rather than weakened a claim.

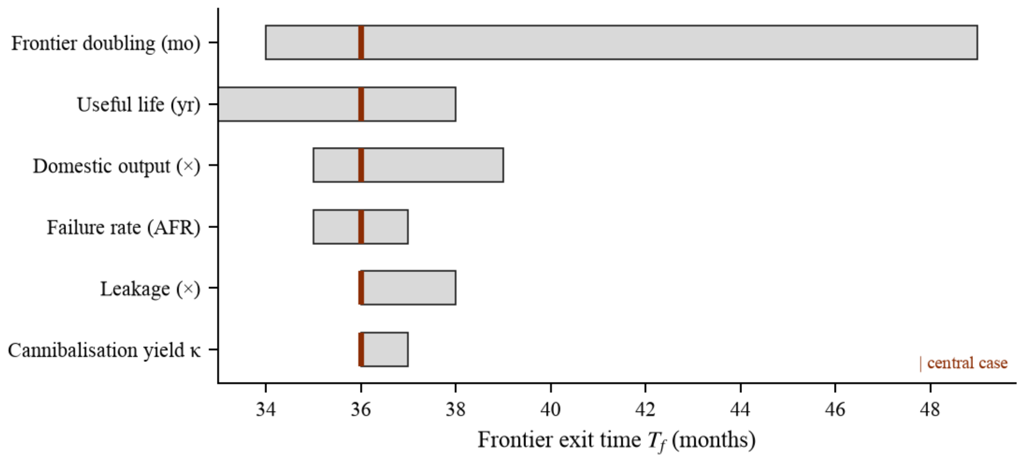

Figure 5. One-at-a-time sensitivity of frontier exit time, ordered by span. The rate of frontier advance dominates; cannibalisation yield and failure rate matter least once the regime is fixed.

The threshold $\theta = 0.10$ is a convention, and its influence is fully transparent: it enters Eq. 5 only through the logarithm, so halving it lengthens every T_f by exactly one doubling time, about five months, and doubling it shortens every T_f by the same amount, leaving ordering and spreads untouched. Verified numerically: 42 / 36 / 31 months for $\theta = 0.05$ / 0.10 / 0.20 in the China central case.

5.4 Regime dependence

Figure 6 shows trajectories across cannibalisation regimes. Values of κ above $\kappa^* = 0.38$, including the aviation value of 10, a hypothetical field-repairable accelerator at 1, and the upper bound of the plausible accelerator range at 0.5, are indistinguishable, because all three are past saturation and the failure flow is already fully absorbed. Only the lower bound, 0.05, sits far enough below the threshold to add materially to the decline. The system does not respond smoothly to κ: it saturates.

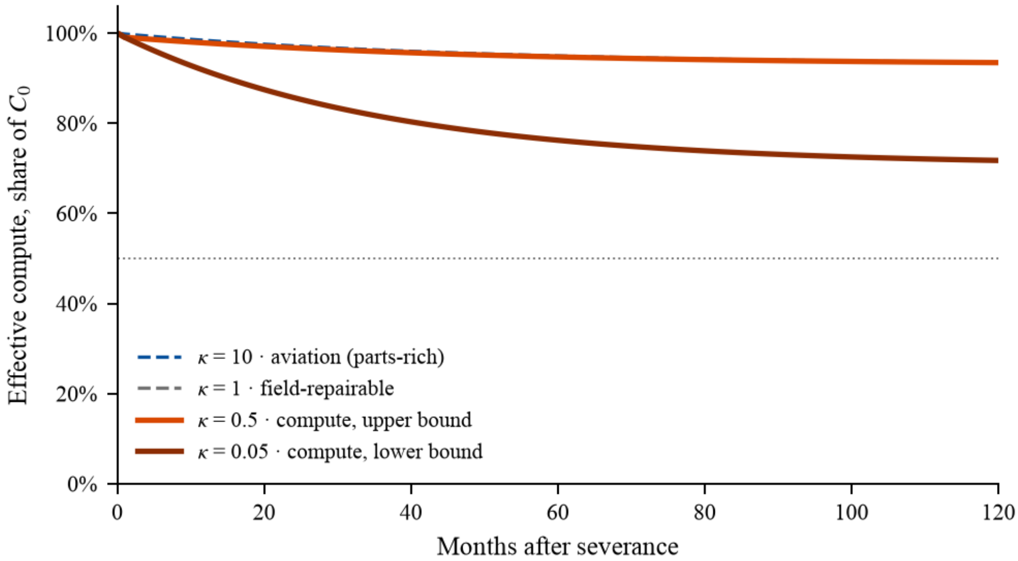

Figure 6. Decay under four cannibalisation regimes. The three curves at or above the critical ratio coincide because all are past saturation, the failure flow is already fully absorbed and additional salvage capability sits idle.

5.5 Partial severance: what leakage actually buys

Real controls are neither complete nor instantaneous. They leak, through diversion, resale and third-country transhipment, and the size of that leakage is the quantity enforcement policy actually contests. Sweeping the leakage term across four orders of magnitude separates two effects that are usually discussed as one.

The two horizons respond to leakage with very different elasticities. Across a sweep from zero to 200% of the installed base per year, the sovereign floor rises from 64% to 803% of C_0, a factor of 12, while frontier exit moves only from 35 to 50 months, a 43% change. The asymmetry follows directly from the analytic form: the floor is an inflow-determined steady state and scales linearly in inflow, whereas T_f enters through a logarithm and through a denominator growing exponentially. Reaching a floor of 100% of C_0 costs about 10% of the installed base per year and buys 2 additional months at the frontier.

Table 3. Partial severance. Effect of leakage on the two horizons.

Leakage regime	H100e / month	% of C_0 per year	T_f (mo)	C_∞ (% of C_0)
None	0	0	35	64
Floor held at 100%	17,253	10.5	37	100
Open-market equivalent	,	200	50	803

Leakage rates are expressed relative to the historically estimated diversion flow. The floor is highly elastic to leakage; the frontier horizon is not.

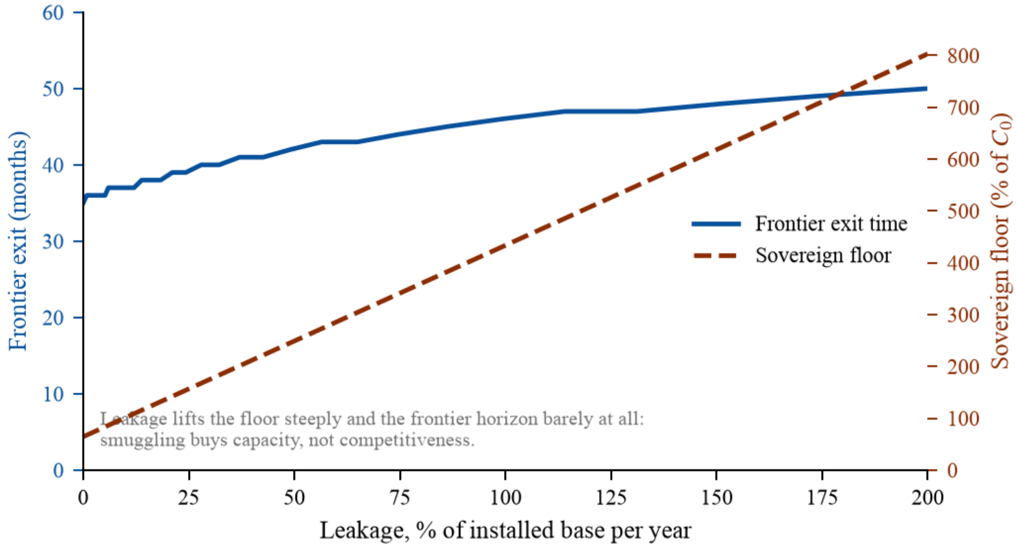

Figure 7. Leakage lifts the sovereign floor steeply while moving the frontier horizon very little. The two curves diverge because the floor is an inflow-determined steady state, whereas frontier exit is governed by the frontier's own growth rate.

Leakage buys capacity, not competitiveness. For enforcement design this inverts the usual emphasis. Interdiction is comparatively effective at protecting a frontier lead, because that lead is set by the frontier's growth rate and by a logarithm of the adversary's stock, both of which resist additional inflow. It is close to powerless at preventing an adversary from maintaining a substantial sustained capacity, because the floor responds to inflow linearly and the required inflow is modest. A control regime justified on the first ground and evaluated on the second will appear to be failing when it is not, and the reverse.

6 The aviation comparison, and why calibration fails

A severance model invites the objection that it is unfalsifiable, since no jurisdiction has yet been fully severed from AI compute. The natural response is to calibrate against the nearest observed case. Russian civil aviation after February 2022 is the only documented severance of a complex, spare-parts-dependent capital fleet with a multi-year observation window: a starting fleet of roughly 1,500-1,800 Western-built aircraft, more than a third reported cannibalised by October 2025, a projected reduction exceeding 50% by 2026, and a documented practice of dismantling approximately one airframe to keep ten flying.

We attempted this calibration and it fails. Fitting the model's hazard as a single free parameter with $\kappa = 10$ fixed from observed practice, the best attainable fit reaches a root-mean-square error of 0.222 and leaves 34 percentage points of the observed decline unexplained at month 47, with the fitted hazard pinned against the top of the search grid.

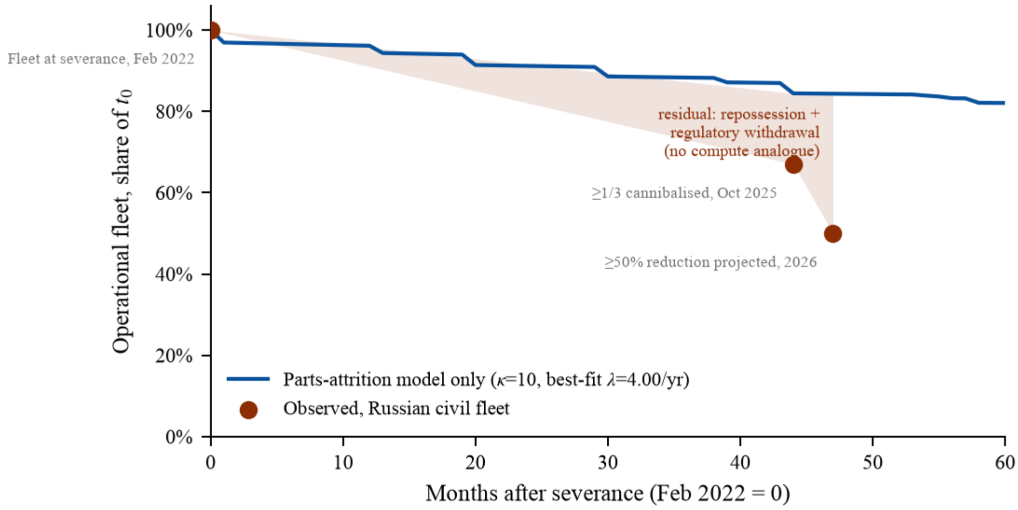

Figure 8. Observed Russian civil fleet against the best attainable parts-attrition-only fit. The shaded residual is the share of decline that parts attrition cannot account for. It is not model error: it corresponds to mechanisms with no compute analogue.

The residual is informative rather than embarrassing. Two mechanisms drive a large part of the observed decline and have no counterpart in compute: repossession of foreign-owned airframes by lessors, which is a one-off stock shock rather than attrition, and withdrawal of airworthiness certification by regulators, which removes serviceable aircraft for legal rather than physical reasons. Accelerators are owned outright and no regulator grounds a degraded cluster. Attributing the entire Russian decline to parts attrition would overstate the hazard by roughly an order of magnitude.

We therefore use aviation to bound κ and to establish that cannibalisation is a real, observed, quantified mechanism, not to calibrate the compute model. Reporting this negative result matters for two reasons. It prevents a spuriously precise calibration from propagating into the literature, and it identifies exactly what an adequate empirical test would require: telemetry from a fleet under genuine resupply constraint, which does not yet exist in public form.

One qualitative finding does transfer. Reported incident frequency in the Russian fleet more than doubled relative to 2019 well before fleet counts halved. Degradation appears as declining reliability before it appears as declining inventory. If the same holds for compute, effective capacity falls faster than node counts suggest, and our estimates are conservative.

7 Discussion

7.1 What severance actually buys

The results support a reframing of the export-control debate. The relevant quantity is not how much compute an adversary loses but how long it takes them to fall off a moving frontier, and on our estimates that interval is roughly two years and is remarkably insensitive to the size of the installed base. Controls do not need to destroy capacity. They need only hold it constant while the frontier advances. Conversely, a jurisdiction that has secured a large stock of accelerators has bought less strategic time than the stock's size implies, because the binding constraint is the frontier's growth rate rather than its own hardware.

This has a symmetrical and less comfortable implication. If frontier advance slows, through scaling limits, energy constraints or capital withdrawal, the value of controls decays with it. Export controls are, on this reading, a bet on continued exponential progress. They are most effective precisely when they are least needed to establish dominance, and least effective when progress stalls and installed stocks retain their relative worth.

7.2 Sovereignty has a number, and it is low everywhere

The sovereign floor makes explicit something the indexing literature has gestured at without quantifying, but it must be read with a caveat we state before the result rather than after it. Appendix A shows that where domestic output is parameterised as a coverage fraction of installed stock, as it is for the United States and European Union, the floor collapses to $cov\ (1 + min(\kappa,\ \kappa^*))$, independent of C_0, λ and L. The US figure of

31.1% is therefore the assumed 15-35% coverage band rescaled by a factor bounded between 1.0 and 1.5, not an independent finding. We report it as a **scenario mapping**: coverage of 0.15 / 0.25 / 0.35 implies a floor of 19 / 32 / 45%, and readers who dispute the coverage assumption can substitute their own. China's floor (53.4%) is different in kind, it derives from an absolute inflow estimate rather than a coverage fraction, and is the only one of the three that is a model output in the ordinary sense. Its interval must then be taken seriously: the 90% band runs from 22.1% to 155.7% of the installed base, crossing parity at the top. The upper end is driven jointly by the stockpile-dependence parameter, whether Huawei's realised 2025 output could be sustained without the pre-control TSMC die stockpile, and by the wide uncertainty on C_0 itself. The model therefore does not license the claim that China cannot sustain its base domestically. It licenses agnosticism, and it identifies the measurement that would resolve the question, the stockpile-dependence share, as the second most valuable empirical target after κ.

With that qualification the qualitative point stands, and it does not depend on the disputed parameter. Lithography is Dutch, leading-edge fabrication is Taiwanese, and high-bandwidth memory is roughly 90% Korean. No participant in this system is severable from it without cost, and the rhetorical framing in which sovereignty is a property one side possesses and the other lacks is not supported by any coverage assumption we consider plausible.

7.3 Testable predictions

The model is falsifiable, and it is worth being specific about how, because a severance model that cannot be checked is an essay. Four predictions follow directly, each with the observation that would refute it.

- **Reliability degrades before inventory does.** In any compute fleet operating under a genuine resupply constraint, the rate of unrecovered job failures should rise measurably before the operational node count falls appreciably. Refuted by observing node counts falling in proportion to, or ahead of, failure rates.
- **Salvage effort shows a saturation point, not a gradient.** Operators who increase cannibalisation capability beyond the threshold should see no further improvement in fleet availability. Refuted by a smooth, unsaturating relationship between salvage effort and availability.
- **Frontier exit scales logarithmically with stock.** Across jurisdictions, T_f should rise by about $T\_dbl \log_2(10) \approx 17$ months per tenfold increase in installed compute, not proportionally. Refuted by a linear or near-linear relationship. (An earlier draft offered the opposite prediction, that position *decouples* from stock, which under the then-implemented metric was unfalsifiable by construction. Its replacement is a genuine prediction with a defined refutation.)
- **Diversion moves the floor, not the frontier.** Periods of increased diversion should raise a jurisdiction's sustained compute capacity substantially while leaving its largest training runs roughly where the frontier clock puts them. Refuted by frontier-scale runs tracking diversion volumes.

The first is the most immediately checkable: it requires only that one operator under constraint publish job-failure telemetry alongside node counts. No such series is public today, which is why §6 reports a failed calibration rather than a successful one.

7.4 Limitations

- **Jurisdictional attribution.** The best public stock data resolve ownership by firm, not country. We sweep shares rather than assert them, but no treatment fully removes this weakness. European results in particular should be read as scenario illustration.
- **κ is not measured.** The cannibalisation yield is bounded by architectural reasoning, not observation, and the plausible range straddles the critical threshold. This is the single most consequential open parameter.
- **Utilisation is held constant.** Energy and grid constraints are represented by a fixed ceiling. Several recent analyses argue power rather than silicon is now binding in some jurisdictions; endogenising $u(t)$ would likely shorten the horizons reported here.
- **Severance is modelled as complete and instantaneous.** Real controls are partial, phased and leak. The leakage term captures some of this; tiered or escalating regimes are not modelled.

- **The frontier is exogenous.** We treat frontier growth as an external exponential. In reality a severed jurisdiction's exit from the frontier alters competitive dynamics and hence the frontier's own trajectory.
- **No empirical severance case exists for compute.** The model is falsifiable in principle, it predicts specific degradation trajectories, but cannot yet be tested against observation, as §6 establishes.
- **Retirement is an economic choice that the model treats as physics.** The useful life L encodes withdrawal at book life, but an operator under severance would plausibly run hardware to failure instead. Re-running the central cases with L set to forty years raises T_f by only three to four months, lengthens every $T_{1/2}$, and lifts floors substantially: the United States coverage scenario from 32% to 92% of C_0, China from 86% to 246%. Run-to-failure therefore strengthens the sovereignty findings and barely moves the frontier clock, which is why the conservative retirement case is the one reported in the main results.
- **T_f measures compute position, not capability position.** Algorithmic efficiency in language models has historically halved the compute needed for fixed performance roughly every eight months (Ho et al., 2024). To the extent such advances are public, both sides gain them and the relative ordering is preserved, but a frozen fleet keeps gaining absolute capability, so crossing θ is softer in capability terms than in FLOP terms. To the extent frontier advances stay proprietary, the severed side degrades faster than T_f suggests. The measure is exact about hardware and deliberately agnostic about software.
- **Remote access is outside the model by construction.** $N(t)$ counts accelerators physically installed and operated within the jurisdiction. Compute rented from foreign-hosted providers is not in $N(t)$, and, unlike the leakage term, whose units enter the fleet and thereafter decay at λ and δ, leased capacity is replenished by the lessor and does not decay at all. Every horizon reported here is therefore conditional on hardware severance *and* effective interdiction of remote access; against hardware-only severance they are upper bounds on the effect. One consolation follows analytically: because the frontier deflator is exponential, a fixed lease of any size only shifts T_f by $T\_dbl \log_2(1 + A/C_0)$. Leased capacity defeats the mechanism only if it compounds at the frontier rate itself, $g^* = \ln 2 / T\_dbl \approx 13\%$ per month.
- **Domestic output is frozen at its severance-date rate.** The model contains no induced-substitution channel, so C_∞ is defined conditional on zero indigenisation growth; for any positive growth rate there is no asymptote at all. This assumption is contradicted by the most recent observed trend in the series the parameter is calibrated from, reported Chinese accelerator output targets roughly doubled between 2025 and 2026, though under the paper's own complete-severance premise, with Korean memory and Dutch lithography both withdrawn, the growth rate could plausibly take either sign. T_f is robust to this: it moves by only a few months across growth rates from *-30%* to *+60%* per year, because no plausible indigenisation rate competes with a frontier that doubles every 5.2 months.

8 Conclusion

National AI capacity has been measured as a stock at least eleven times and modelled as a flow not once. We have argued that the flow is the policy-relevant quantity, built a decay model of installed accelerator fleets under severance, parameterised it from published telemetry rather than assumption, and reported three quantities intended to be reused: the compute half-life, the frontier exit time, and the sovereign floor.

Four findings stand out. Salvage absorbs the failure flow only up to a saturation ratio $\kappa^* = \lambda L \approx 0.38$, and because high-bandwidth memory is co-packaged with the die the plausible range for accelerators straddles that threshold, making κ the field's most valuable unmeasured parameter. Frontier exit occurs at two to four years and rises only logarithmically with installed stock: a tenfold larger fleet buys roughly seventeen further months, and hardware decay accounts for just 13-14% of the interval, the frontier clock for the rest. At the median no major jurisdiction appears self-sufficient, though for the United States and European Union that number is a coverage assumption rescaled rather than an independent result, and China's interval is wide enough to cross parity, leaving the sovereignty question there genuinely open. And leakage, when partial, buys capacity rather than competitiveness: it lifts the sovereign floor steeply while leaving the frontier horizon almost where it was.

The claim that export controls buy time is, on this analysis, correct in form and materially wrong in the mechanism usually assumed. What is bought is not the degradation of an adversary's hardware but the freezing of its position against a frontier that keeps moving. That is a different policy instrument with a different expiry date, and it should be argued for on those terms.

Appendix A Derivation of the critical ratio

Let P_t be the salvage pool. Over one period the pool gains κ repairs for every unit leaving service, unrepaired failures $(F_t$-$\rho_t)$ and retirements Ret_t, and loses ρ_t to repairs performed. Consider the steady state in which every failure is repaired, so $\rho_t = F_t = N\lambda$ and $Ret_t = N\delta$. The pool balance is

$$\Delta P = \kappa N\delta - N\lambda \qquad \text{(A.1)}$$

which is non-negative precisely when $\kappa \geq \lambda/\delta$. Writing $\delta = 1/L$ gives $\kappa^* = \lambda\ L$. Fleet size N cancels, so the threshold is scale-free; inflow R does not appear, so it is also policy-independent. Below κ^* the pool is exhausted in finite time and unrepaired failures accumulate; above it the pool grows without bound and the repair constraint ceases to bind, which is why trajectories for $\kappa = 0.5$, 1 and 10 coincide in Figure 6.

Note carefully what this does *not* establish. Equation A.1 governs the repair constraint alone. Retirement removes units from service whether or not failures are repaired, so N declines above the threshold as well, at rate δ, rather than at δ plus an unabsorbed-failure term. An earlier draft of this paper stated that the installed base *persists* above κ^*; direct simulation refutes that, and the corrected reading, a saturation threshold separating failure-limited from obsolescence-limited decay, is the one verified numerically in §5.1.

A.2 Closed form for the sovereign floor

An earlier draft asserted that no closed form existed and reported the floor from a 900-month integration. One does exist. Setting the pool stationary in Eq. 2 gives $\rho = \kappa(F+Ret)/(1+\kappa)$, hence a net attrition of $(\lambda+\delta)N/(1+min(\kappa,\kappa^*))$ and

$$\frac{C_\infty}{C_0} = \frac{R_{\text{dom}} + R_{\text{leak}}}{C_0\,(\lambda+\delta)}\left(1+\min(\kappa,\kappa^*)\right) \qquad \text{(A.2)}$$

which reproduces the integrated values exactly. The consequence must be stated plainly because it constrains how two of our three floors may be read. Where domestic output is parameterised as a coverage fraction of installed stock, $R\_dom = cov\cdot C_0(\lambda+\delta)$, which is how we treat the United States and European Union in the absence of jurisdiction-resolved production data, this collapses to

$$\frac{C_\infty}{C_0} = \text{cov}\,(1+\min(\kappa,\lambda L)) \qquad \text{(A.3)}$$

independent of C_0, λ and L, with the bracket bounded in *[1.0, 1.5]* over all parameter values we consider. Those two floors are therefore a rescaling of an assumption, not an emergent property of the model, and §7.2 reports them as a scenario mapping. China's floor, which derives from an absolute inflow estimate rather than a coverage fraction, is not subject to this collapse.

Appendix B Reproducibility

All results derive from a single deterministic script with a fixed random seed (20260808). Running it regenerates every figure, table and numerical claim in this paper, including the values quoted in the abstract. The model is 200 lines of NumPy with no external solver dependency. Model code, parameter file, Monte Carlo output and the full source-retrieval log are archived with this preprint; see *Data and code availability*.

Data and code availability

Model code, parameter definitions with per-value source attribution, Monte Carlo output, processed tables and all figure-generating code are openly available at github.com/diShine-digital-agency/The-Half-Life-of-Compute and are archived with the Zenodo record of this article. Underlying data are third-party and publicly available: Epoch AI publishes accelerator ownership and sales data with documented methodology; failure statistics are drawn from published papers. No proprietary or licensed data were used.

Declarations

Funding. This research received no external funding. **Competing interests.** The author is founder of diShine, a consultancy working on AI adoption and governance, and is the author of a trade book on the geopolitics of technology cited once in this paper for a framing contribution. Neither relationship influenced the analysis, which rests entirely on public data and released code. **AI assistance.** Computational tooling was used for source retrieval, model implementation and manuscript preparation; all analytical claims, parameter choices and interpretations are the author's, and every factual claim was verified against a retrieved primary source.

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